Intersecting Inequalities:

Mapping Energy Poverty and Women's Empowerment Across India

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Abstract

Energy challenges in India are particularly acute, with energy poverty affecting households in multidimensional ways. Limited access to affordable and clean cooking fuels, combined with constrained women's agency, exacerbates time poverty and health risks and restricts women's participation in paid employment, undermines women's empowerment. While existing studies show a close link between energy access and gender outcomes in the Indian context; however, substantial regional disparities persist. This study examines the relationship between household energy access and women's empowerment in India at national and state levels using data from the National Family Health Survey (NFHS-5). The study employs descriptive statistical methods and the Concentration Index (CI) to assess inequalities in access to clean cooking fuel and electricity and their association with key empowerment indicators. The findings reveal pronounced interstate disparities, with southern and western states showing higher energy access, while central and eastern states continue to lag. Improved energy access is associated with greater women's decision-making autonomy, higher participation in paid employment, and improved access to financial and digital resources. The results underscore the need for state-specific policies to expand clean energy access in lagging regions and strengthen women's economic empowerment.

Keywords: Energy Poverty, Women's Empowerment, Energy Access, Inequalities.

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INTRODUCTION

Energy poverty in India remains multidimensional, encompassing incomplete electrification, low adoption of clean cooking fuels, unreliable supply, and intra-household constraints on women's control of appliances (Khanna R. A., Lib, Mhaisalkarc, Kumar and Jia Liangd, 2019; Abbas, Khizar, Shixiang, Li., Xu, Baz, and Rakhmetova, 2020; Rafi, Naseef, Prasad, 2021). These deficits increase women's time burdens, heighten exposure to household air pollution, and restrict opportunities for paid employment, information access, and participation in decision-making. NFHS-5 reports improvements in indicators such as bank account ownership, mobile phone access, and menstrual hygiene, yet disparities persist across states and between rural and urban areas. Household power relations strongly influence energy adoption (Pachauri and Rao, 2013), and unreliable services slow progress in empowerment outcomes. Energy access and empowerment are deeply interconnected, with deficiencies in one domain reinforcing disadvantages in the other (Ghosh, Chatterjee, Paul, Ghosh and Husain, 2024; Kelkar, Nathan, Rengalakshmi, Menon, Gaikwad, Shakya, Shrestha & Pakhrin Tamang, 2019). Energy poverty is not solely technical or infrastructural (Munro, Paul, Horst, Healy, 2017) but embedded in social, economic, and gendered structures that shape resource allocation and household decision-making (Parikh, 1995). Women's limited agency over energy choices (Tjøring, Lise, Lynge Jensen, Gårn Hansen, Mørch Andersen, 2018; Kelkar *et al.*, 2019) reflects entrenched norms around asset ownership, mobility, and financial control, constraining their capacity to benefit from modern energy even when nominally available.

Despite growing attention to energy transitions and gender outcomes, most studies examine energy poverty and women's empowerment separately or treat gender as secondary. Empirical work systematically mapping their intersection across Indian states with comparable indicators remains limited. Addressing this gap is crucial to explain why improved energy infrastructure does not consistently yield gains in women's well-being and agency. India's inter-state diversity further complicates the relationship, as states vary in economic development, policy capacity, social norms, and gender outcomes. Some states show convergence between energy access and women's agency, while others exhibit overlapping deprivations (Nandwani and Jain, 2024), especially in rural and marginalized communities (Dutta and Sahu, 2025). Energy poverty is higher in economically backward states, reflecting uneven development (Rafi *et al.*, 2021). National averages thus conceal spatial inequalities and fail to capture how energy poverty and gender disadvantage intersect at the sub-national level (Dutta and Sahu, 2025).

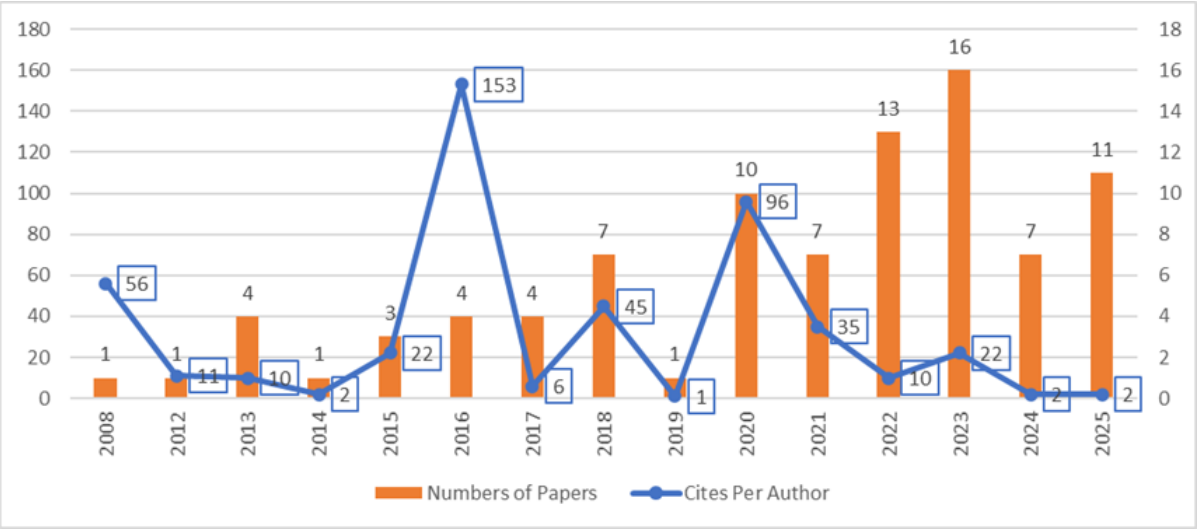
In this background the objective of the present study is to assess the dynamics of energy poverty and women's empowerment in India (national and state-specific), using NFHS data (NFHS 5: 2019-21) to examine associations between household energy access and multiple empowerment outcomes and how these associations vary across states. The research aligns with and contributes to several Sustainable Development Goals (SDGs), including Goal 5 (Gender Equality), Goal 7 (Affordable and Clean Energy), Goal 3 (Good Health and Well-being), and Goal 13 (Climate Action). By identifying empowerment-led pathways to eradicate energy poverty, the current study supports a broader vision of inclusive and sustainable development.

The paper is organised as follows. Section 2 discusses the existing body of literature with a focus on conceptualization, and key thematics covered. The following section provides the description of the data and methods. The findings are described in section 4. The final section provides concluding remarks and throws some light on the intended future works.

ENERGY ACCESS AND GENDER EMPOWERMENT: A LITERATURE REVIEW

The present section employs bibliographic meta-data analysis (2008–2025) to examine the intersection of energy poverty and women's empowerment across Indian states. An initial search yielded 2,000 documents; after exclusions for missing author names (247), incomplete titles (2), non–peer-reviewed sources (983), absent abstracts (215), and lack of relevant keywords (462), a final sample of 90 research articles was selected. The data were analysed using VOS viewer to map keyword co-occurrence and thematic clustering, alongside year-wise publication and citation trends to trace the literature's evolution.

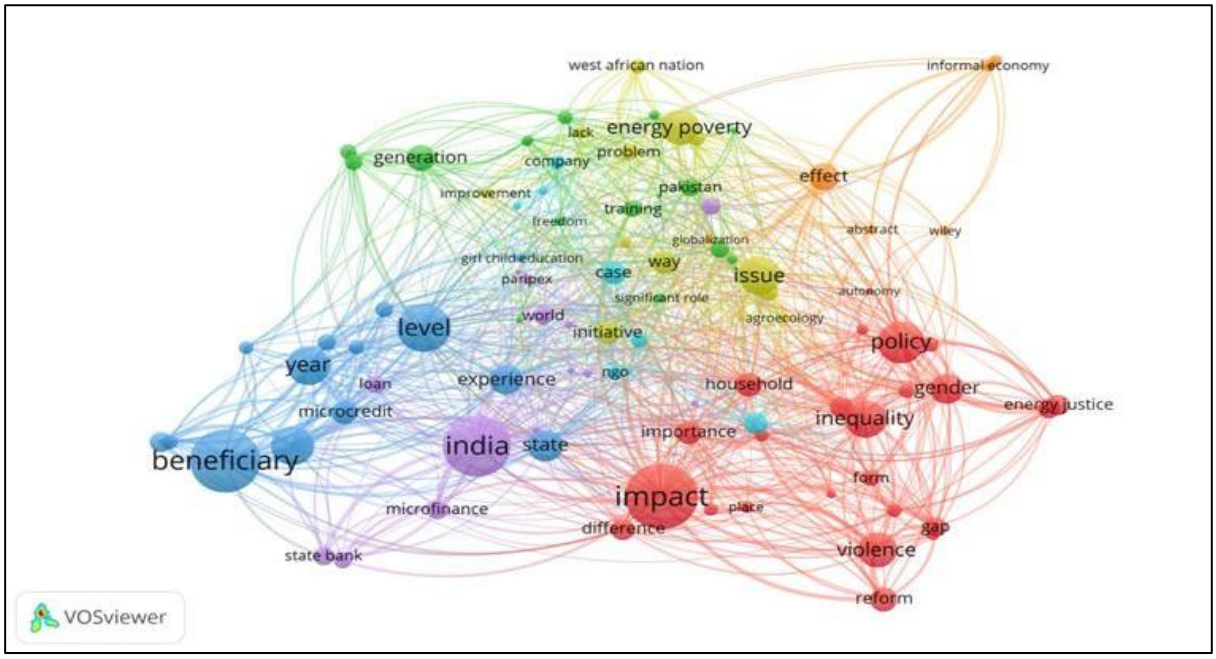
Figure 1: Year-wise Publication and Citation Trends



Source: Author's compilation based on bibliographic metadata of selected research articles (2008–2025).

The temporal distribution of publications shows a steady rise over time. Output was limited between 2008 and 2014, but growth accelerated from 2018, peaking in 2023. Citation impact is higher for earlier studies, especially those from 2016 and 2020, reflecting their foundational role. Recent works add substantially to the literature, though their citation counts remain lower due to limited time for accumulation.

Figure 2: Keyword Co-occurrence Analysis



Source: Author's analysis using VOSviewer based on bibliographic metadata of selected journal articles.

The keyword co-occurrence analysis identifies several thematic clusters in the literature. One cluster highlight gender-related issues such as inequality, policy, violence, and energy justice, underscoring the social and institutional dimensions of energy poverty. Another emphasizes beneficiaries, microfinance, credit access, and household experiences, pointing to the role of financial inclusion in empowerment. A geographically focused cluster centres on India and state-level analysis, reflecting the concentration of empirical studies in this context. Core energy themes as energy poverty, electricity generation, lack of access, and infrastructural challenges form another cluster. In summary, the network shows that research on energy poverty and women's empowerment is multidimensional, spanning social, economic, policy, and infrastructural perspectives.

2.1. Conceptualizing Energy Poverty

Many people, especially in rural areas, remain affected by unclean energy sources. The International Energy Agency (2017, via Abbas *et al.*, 2020) reports 2.8 million premature deaths annually from indoor air pollution, three-fourths among women and children. Energy poverty undermines wellbeing, particularly women's health, with chronic respiratory diseases contributing to premature mortality (Choragudi, 2024). It refers to households lacking affordable, reliable, sufficient, and sustainable modern energy services (Rafi *et al.*, 2021). Those relying on solid fuels like biomass and coal are considered energy poor (Day, Walker, and Simcock, 2016). The UNDP defines it as inadequate access to safe, affordable, and reliable energy for basic needs and economic activity. Similarly, IEA (2010) and ADB (2007) highlight limited electricity and clean cooking fuels, forcing reliance on biomass and restricting livelihoods (Abdoulaye and Lamia, 2022). Rising demand and population growth further limit affordability, excluding many from modern energy services (Abbas et al., 2020). Reductions in energy poverty improve health and education, with electricity access exerting stronger influence than energy use (Banerjee, Mishra and Asfaw Maruta, 2021).

Several studies (Khanna *et al.*, 2019; Abbas *et al.*, 2020; Rafi *et al.*, 2021) conceptualize energy poverty as multidimensional rather than a simple lack of electricity or income. Indices such as MEPI and CEPI (Abbas *et al.*, 2020;

Khanna *et al.*, 2019; Rafi *et al.*, 2021) integrate cooking fuel, indoor air quality, electricity, appliances, and communication services, combining access, affordability, reliability, and quality. Findings show households may remain energy poor despite grid connection if supply is unreliable, fuels unaffordable, or usage insufficient. Rural, poorer, less educated, and female-headed households face higher deprivation, revealing hidden energy poverty obscured by conventional connectivity statistics.

2.2. Women's Empowerment: Multidimensional Frameworks

Empowerment refers to expanding people's ability to make strategic life choices previously denied to them, such as livelihood, residence, spouse, and family size (Kabeer, 1999). It is a process of change through which individuals gain agency and control. Women's empowerment entails decision-making over their own lives, access to opportunities and resources, and participation in societal choices (Odo *et al.*, 2021 via Ghosh *et al.*, 2024). Kabeer (1999) outlines a three-dimensional framework—resources, agency, and achievements emphasizing that resources alone do not ensure empowerment without control and decision-making power. This contextual and multidimensional nature complicates measurement. Supporting this, Ghosh *et al.* (2024) and Chen, Liao, and Zhang (2024) show that empowered women are more likely to live in households using clean fuels and modern energy, indicating infrastructure alone is insufficient without bargaining power.

European studies (Räty and Carlsson Kanyama, 2010; Rainisio *et al.*, 2022; Tjøring *et al.*, 2018; Toro, 2024) add dimensions of self-efficacy in energy saving, responsibility for practices, and breadwinner status, showing that even with stronger formal equality, norms and labour divisions shape energy management and income control. The report *A Framework for Women-Led Development* (Vemireddy, 2025) argues empowerment is socially embedded, linked to familial support (Nayakaratne and Perera, 2024), cultural norms, and structural conditions. Indian studies (Bhattacharya and Banerjee, 2013) highlight that autonomy alone is insufficient without physical well-being and education. Thus, empowerment is best understood as enhancing interrelated capabilities shaped by social and contextual factors. Gressel *et al.* (2020) view it as reducing vulnerability by expanding choices within specific contexts, aimed at sustainability and resilience. Complementary to this, Akter and Chindarkar (2020) describe diffusion of empowerment, where gains spill over to other women in households or family networks over time.

2.3. Intersecting Inequalities of Gender and Energy: Linkages in the Literature

The literature highlights a strong link between gender inequality and energy poverty, where norms heighten women's vulnerability and limited energy access constrains their opportunities. Women, as primary energy users, spend more time collecting biomass (Acheampong, Osei Opoku, Amankwaa and Dzator, 2024) and bear disproportionate burdens due to labour divisions and institutional barriers (Dutta, Soma, Kooijman and Cecelski, 2017). Gender-blind policies worsen these inequalities, while gender-responsive interventions improve health, time use, and empowerment. Household chores expose women to unclean cooking fuels, causing health risks, especially in rural areas (Cicowiez *et al.*, 2022; Pradhan, Mainali, Mokhtara, Lohani, 2025). Access to clean energy enhances equality and education (Acheampong *et al.*, 2024).

Empirical studies show gendered impacts of energy poverty through time burdens, indoor pollution, and constraints on children's health and education (Abbas *et al.*, 2020; Rafi *et al.*, 2021; Kelkar *et al.*, 2019; Acheampong *et al.*, 2024). Children near kitchens face respiratory problems and poor learning outcomes (Das *et al.*, 2025). Women's time poverty limits productive work, with Sub-Saharan women spending three times longer than men on unpaid labour (Jagoea *et al.*, 2020), while globally women average 4.5 hours daily (Rodgers, 2024). Clean cooking transitions reduce cooking and

fuel collection by nearly an hour (Maji, Mehrabi and Kandlikar, 2021). Rural electrification lowers unpaid work and raises market participation, though gains remain limited without broader measures (Cicowiez *et al.*, 2022). Yet, interventions ignoring women's priorities risk new injustices (Munro *et al.*, 2017; Mang-Benza *et al.*, 2023; Okyere and Boqiang, 2023). Female-headed households with disabilities face particularly high deprivation (Okyere and Lin, 2023).

Women act as central energy managers yet remain marginal decision makers, constrained by social norms. Ghosh *et al.* (2024) and Chen *et al.* (2024) show that adoption of clean fuels and modern energy increases women's participation in household financial and purchase decisions, particularly in urban, wealthier, and educated households, though gaps persist for poorer rural women. In Nigeria, women are solely responsible for fuelwood collection, processing, and cooking, accounting for nearly 60 percent of household energy demand, and they make over 62 percent of cooking-energy decisions (Yahaya, Nabinta and Olajide, 2007).

Qualitative studies in Europe reveal similar dynamics: women manage daily adjustments for electricity use and energy saving, while men dominate technical and strategic decisions (Tjøring *et al.*, 2018; Rainisio *et al.*, 2022). This echoes Parikh's (1995) observation that women bear the burdens of energy poverty without equivalent influence over technologies or investments. These findings suggest that household, community, and policy decision spaces must be analysed together to understand how intersecting inequalities shape women's energy experiences.

Table 1: Impact of Energy Poverty

Impact of Energy Poverty	**Brief Description**	**References**
Time poverty and unpaid work burden	Women spend long hours collecting traditional fuels and managing low-quality energy, reducing time for paid work, rest and education.	Kelkar *et al*. 2019; Cicowiez *et al.* 2022; Munro *et al*. 2017; Jagoe *et al*., 2020; Dutta and Sahu, 2025; Acheampong *et al*., 2024; Rodgers, 2024.
Adverse health outcomes and indoor air pollution	Reliance on solid fuels and smoky kitchens increases respiratory problems, undernutrition, stunting and other health risks, especially for women and children.	Rafi *et al.* 2021; Abbas *et al.* 2020; Kelkar *et al*. 2019; Choragudi, 2024; Pradhan et al., 2025.
Reduced educational enrolment and learning	Children, especially girl children and women in energy-poor households are less likely to be enrolled in school and perform worse in reading, writing and maths due to time constraints and poor study conditions.	Rafi *et al.* 2021; Acheampong *et al*. 2024; Pradhan et al., 2025.
Constraints on women's empowerment and agency	Limited access to modern energy restricts women's mobility, decision making power and participation in income-generating activities.	Kelkar *et al.* 2019; Ghosh *et al.* 2024; Chen *et al.* 2024; Choragudi, 2024.
Lower labour market participation and incomes	Lack of reliable electricity and modern energy reduces opportunities for productive work and enterprise, particularly in rural areas.	Cicowiez *et al*. 2022; Abbas *et al.* 2020; Choragudi, 2024; Pradhan et al., 2025
Intergenerational human capital losses	Persistent energy deprivation generates long-term deficits in child health and education, constraining human capital formation across generations.	Rafi *et al.* 2021; Kelkar *et al*., 2019; Das *et al*. 2025.
Unequal distribution of benefits from energy interventions	Modern energy projects sometimes bypass women or traditional users, reinforcing distributive and recognition injustices.	Munro *et al.* 2017; Mang-Benza *et al.* 2023; Murshed, 2024.

Source: Authors compilation.

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2.4. Policies and Programs Addressing Energy Poverty and Gender in India

India has introduced several national policies to tackle energy poverty and gender inequality (Table 2), with mixed success in achieving comprehensive access and gender-sensitive outcomes. Programs such as PMUY, Saubhagya, JNNSM, DDUGJY, NBMMP, and NPIC have expanded clean cooking fuel and electricity access, reducing indoor air pollution and women's fuel collection burdens. These initiatives have improved health outcomes, lowered respiratory

illnesses, and supported women's labor participation. PMUY's LPG distribution reduced drudgery, while Saubhagya's electrification displaced kerosene lamps, enabling extended study hours and small enterprises. JNNSM accelerated solar deployment, DDUGJY strengthened rural infrastructure, and NBMMP/NPIC promoted biogas and efficient stoves to curb emissions and biomass dependence. However, sustainability challenges persist: PMUY faces low refill rates due to cost and cultural barriers; Saubhagya and DDUGJY struggle with unreliable supply and DISCOM losses; JNNSM suffers quality and funding issues; IEP remains coal-dependent; NPIC lacks subsidies; and smaller solar initiatives scale unevenly.

Table 2: List of Programmes Addressing Energy Poverty and Gender in India

Policy/Program	Year of Implementation	Features
National Programme on Improved Chulhas (NPIC)	1985 (discontinued centrally in 2002)	Distributes efficient biomass stoves to reduce fuel consumption and indoor pollution; transitioned to state-level implementation.
National Biogas and Manure Management Programme (NBMMP)	2002	Subsidizes family and community biogas plants using animal dung for clean cooking fuel and manure, with technical support and evaluations.
Integrated Energy Policy (IEP)	2006	Comprehensive framework for energy security, including clean fuel access (LPG, biogas) for all households and fuelwood plantations near villages; encourages women's self-help groups for biofuel management.
Light a Billion Lives (LaBL) by TERI	2008	Provides solar lighting and cooking systems; trains women from self-help groups as energy entrepreneurs for procurement, installation, and maintenance.
Jawaharlal Nehru National Solar Mission (JNNSM)	2010	Promotes solar energy deployment, including off-grid systems and community projects targeting 100 GW capacity, with emphasis on rural energy access.
Deen Dayal Upadhyaya Gram Jyoti Yojana (DDUGJY)	2014	Flagship rural electrification scheme aimed at feeder separation, strengthening sub-transmission and distribution networks and providing continuous power to rural households.
Pradhan Mantri Ujjwala Yojana (PMUY)	2016	Provides free LPG connections and subsidies for refills to below-poverty-line women households to shift from biomass to clean cooking fuel, reducing health risks and time spent on fuel collection.
Pradhan Mantri Sahaj Bijli Har Ghar Yojana (Saubhagya)	2017	Aims for universal household electrification through grid and off-grid solutions, improving access to lighting and appliances in rural areas and urban poor to alleviate energy poverty.
Hariyali Gram Initiative	2022	Women-led decentralized renewable energy solutions in rural villages, focusing on carbon reduction, energy security, and livelihood generation through policy convergence.

Source: Authors compilation from Ministry of Power, Ministry of New & Renewable Energy and Government of India.

Policy literature argues that tackling energy poverty and advancing gender equality requires gender-responsive, institutionally coherent interventions rather than access-only schemes. Dutta and Sahu (2025) highlight the limits of India's energy policies, noting that rural electrification reduces health risks but fails to address women's disproportionate burden from cooking energy deprivation. Clean cooking programmes yield gender benefits only with exclusive LPG use, while partial adoption and fuel stacking perpetuate risks (Nandwani and Jain, 2024). Lessons from NPIC stress demand-side reforms (Hanbar and Karve, 2002), and affordability inequities remain critical (Asharaf and Richard, 2024; Maji *et al.*, 2021). Historically, macro energy policies treated gender narrowly (Parikh, 1995), and weak coordination undermines inclusion (Buchy and Shakya, 2024). Evidence shows subsidies modestly improve women's outcomes, with greater gains when paired with productivity policies (Cicowiez *et al.*, 2022) or political participation (Han *et al.*, 2025). Thus, successful energy policy must integrate gender-sensitive design, women's representation in decision-making, and complementary livelihood investments to overcome intersecting inequalities in energy poverty and women's empowerment.

In a nutshell, Indian energy programmes, PMUY, Saubhagya, JNNSM, DDUGJY and others have expanded electricity and clean cooking fuel access, improving health, reducing women's drudgery and supporting income-generating activities. Complimentary to this, Javeed (2024) argued that micro credit programmes improve women economic activity and therefore increasing decision making power in the household. However, partial adoption, fuel stacking, affordability challenges and weak gender mainstreaming limit sustainable empowerment outcomes, highlighting the need for coordinated, gender-sensitive, and context-specific energy policies that address intersecting inequalities in energy and gender.

DATA AND METHODS

3.1 Data

The present study is based on the National Family Health Survey (NFHS-5, 2019–21), which provides nationally representative data covering all states and union territories of India. (Tripathi, Pathak, Lahariya, 2023). This survey covers population, health, nutrition and family welfare including women empowerment. Each round of the NFHS has focused on supplying reliable and essential data on health and family welfare to the Ministry of Health and Family Welfare and other agencies for effective policy formulation and programme implementation and it seeks to generate information on emerging and evolving issues related to health and family welfare in the country. The fifth round (NFHS-5) started in two phases by 2019 to 2021. The NFHS data is used because of the reliability and representativeness (Tripathi *et al*., 2023). Survey uses stratified multistage sampling method and precise state-level estimates disaggregated by wealth, caste, and residence, enabling robust geospatial mappings of women's empowerment against energy poverty metrics like clean fuel access.

The current study focuses on household energy access and women's empowerment outcomes. Household energy access is measured through two key indicators: electricity availability and type of cooking fuel. Both variables are recoded into binary form to facilitate interpretation and inequality measurement. Electricity access is coded as 1 if the household has electricity and 0 otherwise, while clean cooking fuel access is coded as 1 if the household uses electricity, LPG, or biogas, and 0 if the household relies on kerosene, coal, wood, dung, crop residue, or other traditional fuels. Women's empowerment is assessed through multiple indicators, including participation in household decision-making (such as decisions on health care, major purchases, and visiting relatives), autonomy in mobility (ability to go to the market, health facility, or outside the village), control over financial resources (ownership of bank accounts, mobile phones, or

property), and access to health services. These empowerment outcomes are also expressed as binary variables to ensure comparability across states.

Table 3: Variable Used in the Study

Variable	Indicator	Brief Description	Source
Access to clean cooking fuel	Energy Access	Households using cooking fuel	Ashraf & Richard, 2024; Dutta and Sahu, 2025; Nandwani and Jain, 2024; Rafi *et al.* 2021.
Access to electricity	Energy Access	Household's access electricity	Ghosh *et al*., 2024; Rafi *et al*., 2021; Day *et al*., 2016; Kelkar and Yu, 2019.
Women Participation in Decision Making	Social empowerment	Married women (15–49) participating in key household decisions.	Ghosh *et al*., 2024; Kabeer, 1999; Akter & Chindarkar, 2020; Sankhyayan and Dasgipta, 2019; Chen *et al*, 2024
Ownership of House or Land	Asset ownership	Women (15–49) owning a house and/or land alone or jointly.	Ghosh *et al*., 2024; Kabeer, 1999; Bhattacharya and Banerjee, 2012.
Type of Earnings from Women's Work	Economic empowerment	Working women (15–49) receiving cash earnings.	Ghosh *et al*., 2024; Kabeer, 1999; Akter & Chindarkar, 2020.
Ownership of Mobile Phone by Women	Access to Information	Women (15–49) owning and using a mobile phone.	Ghosh *et al*., 2024; Kelkar *et al*., 2019; Gressel *et al*., 2020.
Account in Bank or Other Financial Institution	Financial inclusion	Married women (15–49) having and using their own bank account.	Ghosh *et al*., 2024; Kabeer, 1999; Akter & Chindarkar, 2020.
Women Aged 15–49 Using Hygienic Methods of Protection During Menstruation	Health autonomy	Women (15–49) using hygienic menstrual protection methods.	Ghosh *et al*., 2024; Bhattacharya & Banerjee, 2012; Kabeer, 1999.

Source: Authors' compilation

The study's key variable is access to clean cooking fuel, measured as the share of households using LPG or other modern fuels. This indicator captures household-level energy transition, reflecting both infrastructural availability and women's capacity to adopt and sustain cleaner energy. Women's empowerment is assessed across multiple dimensions: decision-making authority (married women aged 15–49 participating in three household decisions), cash income autonomy (women in the same cohort earning cash in the past year), property ownership (ownership of house/land alone or jointly), financial inclusion (women with a bank or savings account they personally use), and digital inclusion (women owning and using a mobile phone). Together, these measures reflect agency, economic control, asset security, resource management, and communication capacity.

Finally, post-menstrual care is captured through the proportion of women in same age cohort who use hygienic methods of protection during menstruation, serving as a proxy for reproductive health, dignity, and agency. Together, these

variables provide a multidimensional framework for analysing empowerment as both an explanatory factor and a mediator of energy transitions, within the broader discourse on gender equality and energy justice.

3.2. Methods

The empirical strategy adopted in this study proceeds in two stages, combining descriptive statistical analysis and with inequality diagnostics through the Concentration Index (CI). The descriptive statistics are employed to measure the empirical outline of women's empowerment and energy poverty across Indian states and union territories. Empowerment is conceptualized as a multidimensional construct encompassing decision-making authority, mobility, financial autonomy, awareness, and participation in household and community affairs, while energy poverty is captured through indicators of clean cooking fuel adoption, electricity access. For each state, measures of concentration and dispersion, including mean values, standard deviations, and minimum and maximum scores are computed to highlight both average levels of empowerment and the extent of heterogeneity across regions and social groups. Descriptive analysis here is critical not only for situating the data within broader socio-economic contexts but also for motivating the inequality diagnostics that follow. The second stage of analysis applies the Concentration Index (CI), a widely recognized measure in development economics for quantifying socioeconomic inequality in outcomes. The CI captures the extent to which empowerment indicators are disproportionately concentrated among households at different points in the wealth distribution. The CI is defined as,

$$CI = 2/\mu \, COV(y,r) \quad \ldots\ldots(1)$$

Where, (y) denotes the empowerment outcome of interest, (r) represents the fractional rank of women empowerment, and μ is the mean of the outcome.

Since the outcome variables in this study are binary, the Erreygers correction is applied to ensure comparability and interpretability of CI values. The corrected index is given by,

$$CI_E = 4 \cdot \mu \cdot (1-\mu) \cdot CI \quad \ldots\ldots. (2)$$

This adjustment accounts for the bounded nature of binary variables and ensures that the corrected CI remains within the conventional –1 to +1 range. The Erreygers CI is therefore more appropriate for comparing inequality across states and indicators when outcomes are expressed as proportions.

The index ranges from (-1) to (+1), with positive values indicating pro-rich concentration, negative values indicating pro-poor concentration, and values close to zero suggesting relative equality in distribution. In practice, CI values are estimated for each state and for each empowerment dimension, with standard errors computed to assess statistical significance. The interpretation of CI results allows for the identification of states where empowerment benefits are disproportionately captured by wealthier women, where targeted schemes have successfully reached marginalized groups, or where apparent equity masks underlying heterogeneity. By combining descriptive statistics with CI analysis, this methodology captures both the breadth of empowerment outcomes and the depth of inequality patterns, situating empowerment as both an outcome and a mediator in the energy poverty context and recognizing the dynamic and recursive relationship between clean energy access and gender equality.

RESULT AND ANALYSIS

State wise Inequalities: Descriptive Analysis

Electricity access across Indian states and union territories is nearly universal, averaging 98 percent. Coverage is highest in Goa, Delhi, Puducherry, Chandigarh, and Lakshadweep (≈100 percent), while even the lowest states—Uttar Pradesh, Meghalaya, and Assam—exceed 91 percent. Low variance (5.48) and standard deviation (2.34) confirm limited dispersion, though negative skewness (–1.71) and high kurtosis (5.07) indicate a few lagging states pulling the mean downward. This reflects the success of national schemes such as Saubhagya, with residual deprivation concentrated in large, rural, and infrastructurally challenged states.

Clean cooking fuel adoption shows far greater inequality. The mean of 64 percent masks wide disparities, from Delhi (98.9 percent) and Goa (96.5 percent) to Jharkhand (31.9 percent) and Bihar (37.8 percent). High variance (444.7) and standard deviation (21.08) reveal substantial heterogeneity. Unlike electricity, the distribution is flatter (kurtosis 1.67), shaped by affordability, supply chain access, and cultural preferences, leaving rural and disadvantaged regions reliant on biomass.

Women's empowerment indicators also vary sharply. Household decision-making is nearly universal in Nagaland, Mizoram, and Puducherry (>97 percent) but lower in Jammu & Kashmir and Karnataka (~82 percent). Cash income is higher in Telangana, Andhra Pradesh, and Tamil Nadu (~40 percent) but limited in Bihar and Uttar Pradesh (~15 percent). Asset ownership peaks in Ladakh and Jharkhand (>64 percent) but is low in Goa and Delhi (~23 percent). Bank account ownership exceeds 90 percent in Puducherry and Tamil Nadu but remains lower in Nagaland (63.7 percent). Mobile phone use averages 65 percent nationally, ranging from Goa (91.2 percent) to Madhya Pradesh (38.5 percent). Hygienic menstrual protection is nearly universal in Puducherry and Tamil Nadu (>98 percent) but lags in Bihar and Madhya Pradesh (~60 percent).

4.2. Association between Inequalities in Energy Access and Women's Empowerment

The correlation matrix for electricity access (Table A.2) shows modest but significant positive links with women's mobile phone ownership, bank accounts, household decision-making, and hygienic menstrual protection at the 1 percent level. These associations suggest electrification supports enabling conditions for empowerment, though magnitudes are small and income effects negligible. Empowerment indicators themselves—phones, financial inclusion, and assets—are positively correlated, reflecting their interconnectedness.

By contrast, clean cooking fuel (Table A.3) exhibits weaker and mixed associations. It is negatively correlated with mobile phone, bank account, and asset ownership, while positively linked to menstrual hygiene. Correlations with decision-making and cash earnings are weak, indicating adoption alone does not drive agency.

Together, the matrices show that electricity access has more consistent positive associations, while clean fuel use presents weaker or negative economic links. As pairwise correlations establish association but not causality, the study employs the Concentration Index (CI) to assess inequality across wealth quintiles, providing deeper insight into whether energy poverty disproportionately affects poorer households and whether empowerment outcomes are skewed toward wealthier women.

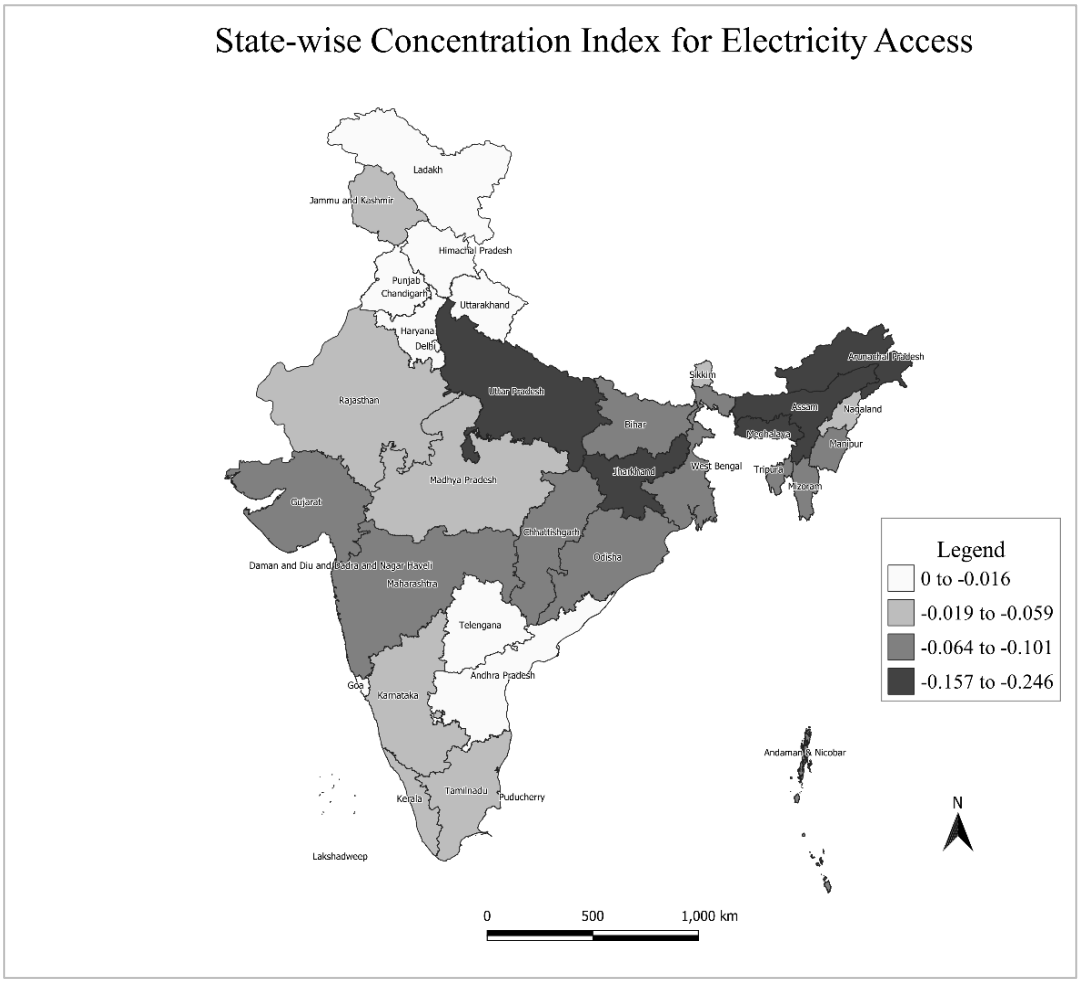
4.3. Concentration between Inequalities in Energy Access and Women's Empowerment

The state-wise results in Table A.4 demonstrate India's remarkable progress toward near-universal electrification, with mean household electricity access exceeding 97 percent in most states and union territories. Several states and UTs, including Goa, Chandigarh, Delhi, Kerala, Tamil Nadu, and Punjab, report electrification rates above 99 percent, reflecting successful implementation of national and state-level electrification programs. However, large states such as Uttar Pradesh (92.1 percent), Assam (93.7 percent), Jharkhand (94.3 percent), and Meghalaya (92.1 percent) continue to exhibit lower average access, suggesting persistent regional disparities. The standard deviations and minimum values further highlight that while most households are connected, pockets of deprivation remain, particularly in socioeconomically diverse and geographically challenging states.

The raw CI values are consistently negative across all states, ranging narrowly between –0.83 and –0.86. This uniformity reflects the fact that households without electricity are disproportionately concentrated among poorer wealth quintiles. In contexts where electrification is nearly universal, the small share of households still lacking access tends to be clustered among the poorest, producing consistently negative CI values. However, because electricity access is a binary variable with very high mean values, the raw CI alone may exaggerate inequality.

The Erreygers-corrected CI provides a more nuanced picture by adjusting for the bounded nature of binary outcomes. Corrected CI values vary considerably across states, from near zero in Goa (0.0000), Chandigarh (–0.0044), and Punjab (–0.0084), to much larger negative values in Uttar Pradesh (–0.2459), Meghalaya (–0.2462), Assam (–0.2002), and Jharkhand (–0.1827). These results indicate that while most states have achieved equitable electricity distribution, a few states continue to exhibit significant inequality, with poorer households disproportionately excluded from access. For example, Uttar Pradesh and Meghalaya stand out as states where electricity deprivation is both more prevalent and more unequally distributed, highlighting persistent regional disparities despite national progress.

Figure 3: State-wise Concentration Index for Electricity Access



Source: *Study area map derived from Survey of India topographical sheets with heat map generated in QGIS software*

The results in Table A.5 highlight stark disparities in clean cooking fuel access across Indian states and union territories. Unlike electricity, which has achieved near-universal coverage, clean fuel adoption remains uneven, with mean values ranging from as low as 26 percent in Meghalaya and 27 percent in Jharkhand to over 95 percent in Goa and 97 percent in Delhi. This wide variation underscores the persistence of energy poverty in several states, particularly in the Hindi heartland (Bihar, Uttar Pradesh, Jharkhand, Madhya Pradesh), the northeastern region (Meghalaya, Nagaland, Assam), and parts of central India (Chhattisgarh, Odisha). States such as Kerala, Karnataka, Maharashtra, and Andhra Pradesh report relatively higher adoption rates (70–80 percent), while union territories like Chandigarh, Puducherry, and Delhi demonstrate near-universal coverage.

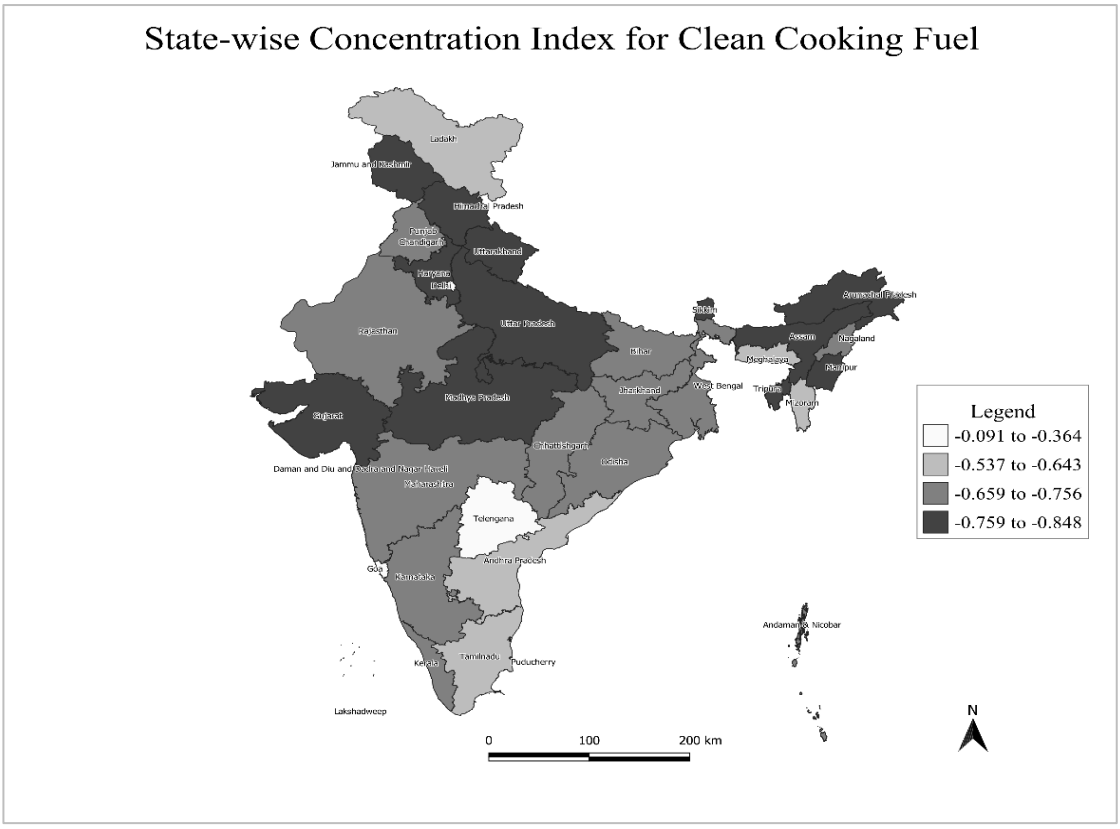
The raw CI values are consistently negative across all states, ranging between –0.81 and –0.85. This indicates that households without clean cooking fuel are disproportionately concentrated among poorer wealth quintiles. However, as with electricity, the raw CI values are compressed and do not fully capture the extent of inequality due to the binary nature of the variable.

The Erreygers-corrected CI provides a more meaningful measure of inequality. Corrected CI values vary substantially across states, reflecting both differences in mean access and the distribution of deprivation. States with very low mean access, such as Himachal Pradesh (46 percent, CI_E= –0.8483), Uttarakhand (42 percent, CI_E = –0.8237), Uttar Pradesh (43 percent, CI_E = –0.8221), and Bihar (32 percent, CI_E = –0.7231), exhibit high negative corrected CI values, indicating that clean fuel deprivation is heavily concentrated among the poorest households. Similarly, northeastern states such as Nagaland (CI_E = –0.7511), Meghalaya (CI_E = –0.6424), and Assam (CI_E = –0.7667) show significant inequality, reflecting both low coverage and pro-poor concentration of deprivation.

By contrast, states and UTs with high mean access report much smaller corrected CI values. Goa (95.7 percent, CI_E = –0.1399), Delhi (97.3 percent, CI_E = –0.0911), Chandigarh (92.8 percent, CI_E = –0.2196), and Puducherry (90.8 percent, CI_E= –0.2850) demonstrate that clean fuel adoption is more equitably distributed across wealth quintiles, with only marginal inequality remaining. Intermediate states such as Kerala (69 percent, CI_E = –0.7231), Maharashtra (71.7 percent, CI_E = –0.6806), and Karnataka (72.4 percent, CI_E= –0.6721) show moderate inequality, reflecting partial progress but continued concentration of deprivation among poorer households.

The analysis shows that while India has made progress in expanding clean cooking fuel access, inequality remains substantial and regionally concentrated. The Erreygers-corrected CI values highlight that the poorest households bear the brunt of deprivation, making clean cooking fuel adoption both an energy poverty issue and a socioeconomic inequality challenge. Addressing these disparities requires targeted interventions—such as subsidies, improved distribution networks, and awareness campaigns—focused on the poorest households and lagging states, to ensure that clean energy transitions contribute to inclusive development and women's empowerment.

Figure 4: State-wise Concentration Index for Clean Cooking Fuel



Source: *Study area map derived from Survey of India topographical sheets with heat map generated in QGIS software*

CONCLUSION

This study demonstrates that persistent inequalities in household energy access in India are closely associated with disparities in women's empowerment across states. The findings show that improved access to clean cooking fuel and electricity is linked to greater women's decision-making autonomy, higher participation in paid employment, and enhanced access to financial and digital resources, with southern and western states performing better than central and eastern regions. These results underscore the importance of state-specific, gender-responsive energy policies to reduce regional disparities and promote women's economic empowerment. The study conceptualises energy poverty not simply as a shortfall in physical energy infrastructure, but as a deprivation of substantive freedoms that limits women's ability to achieve valued functioning as conceptualised by Sen's Capability Approach. Inadequate access to clean cooking fuels and reliable electricity constrains women's capabilities by intensifying time poverty, exposing them to health risks, and restricting their participation in income-generating and decision-making processes. The observed associations between improved energy access and enhanced outcomes in women's autonomy, employment, asset ownership, financial inclusion, and digital connectivity highlight the role of energy as a critical conversion factor that enables resources to be translated into capabilities, particularly in contexts marked by entrenched gender norms. An intersectional reading of the results further reveals that the empowerment effects of energy access are not uniform across social locations. Variations across states reflect the interaction of gender with regional development trajectories, socio-economic status, education, and cultural norms, producing differentiated capability outcomes. The persistence of lagging regions, particularly in central and eastern India, illustrates how multiple and overlapping disadvantages constrain women's ability to convert energy access into meaningful empowerment, even when formal access improves.

The findings challenge technocratic and gender-neutral approaches to energy policy. Given strong interstate variation, state-level prioritisation is necessary (e.g., targeted clean-fuel and menstrual hygiene programs in low-access states; complement electrification with appliance financing in lagging districts). They point to the necessity of embedding energy interventions within a broader framework of gender-just development that explicitly addresses structural and intersectional inequalities. Policy efforts must therefore move beyond access targets to focus on enhancing women's capabilities through complementary investments in childcare, education, skill formation, and labour market integration, particularly in regions characterised by persistent deprivation. Integrating energy programmes with digital literacy and financial inclusion initiatives can further strengthen women's agency by enabling electricity-enabled access to information, banking, and social networks. In this sense, energy policy emerges not merely as an infrastructural concern but as a transformative lever for expanding women's capabilities and advancing gender-equitable development.

From a feminist political economy perspective, the findings underscore how energy deprivation is embedded within gendered structures of unpaid care work, labour market segmentation, and intra-household power relations. The unequal burden of energy scarcity borne by women reinforces their disproportionate responsibility for domestic labour and care, thereby sustaining gendered divisions of labour and limiting economic agency. Improved access to modern energy services appears to partially alleviate these structural constraints by reducing unpaid work burdens and expanding opportunities for women's engagement in paid employment and financial systems. However, the uneven translation of energy access into empowerment outcomes across states suggests that energy transitions are mediated by institutional arrangements, labour market conditions, and state capacity, rather than operating as automatic pathways to gender equality.

This study has the potential to generate actionable insights for policy and program design. Government schemes often adopt a supply-side approach, focusing on affordability and infrastructure. However, without a nuanced understanding of intra-state inequality dynamics, such interventions may fall short of achieving their intended impact. By highlighting the role of women's autonomy in clean fuel adoption, this research can inform more holistic, gender-responsive policies that go beyond provisioning to empowerment.

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Appendix:

Table A.1: Descriptive statistics

	Electricity access	Households using cooking fuel	Married women (15–49) participating in key household decisions	Working women (15–49) receiving cash earnings	Women (15–49) owning a house and/or land alone or jointly	Married women (15–49) having and using their own bank account.	Women (15–49) owning and using a mobile phone	Women (15–49) using hygienic menstrual protection methods
Min. Value	91	31.9	80.4	10.9	15.8	63.7	38.5	58.8
Max Value	100	98.9	99.2	45.1	72.2	92.6	91.2	99.1
Mean	98.04	64.17	90.56	27.17	42.30	79.68	65.09	83.78
Standard Deviation	2.34	21.08	4.40	9.05	17.28	6.94	15.18	11.87
Variance	5.48	444.70	19.44	81.97	298.76	48.17	230.64	140.97
Skewness	-1.71	-0.04	-0.30	0.31	0.14	-0.10	0.05	-0.55
Kurtosis	5.07	1.67	2.97	2.12	1.70	2.55	1.68	2.15
No. of Observation	36	36	36	36	36	36	36	36

Source: Author's estimation based on NFHS 5

Table A.2: Pairwise correlation Matrix of electricity

Variable	Access to Electricity	Ownership of Mobile Phone by Women	Account in Bank or Other Financial Institution	Type of Earnings from Women's Work	Women Participation in Decision Making	Ownership of House or Land	Women (15–49) Using Hygienic Menstrual Methods
Access to Electricity	1.0000						
Ownership of Mobile Phone by Women	0.0512***	1.0000					
Account in Bank or Other Financial Institution	0.0123***	0.2250***	1.0000				
Type of Earnings from Women's Work	-0.0042	0.0332***	0.0285***	1.0000			
Women Participation in Decision Making	0.0030**	0.0690***	0.0438***	0.0191***	1.0000		
Ownership of House or Land	0.0123***	0.2250***	1.0000***	0.0285***	0.0438***	1.0000	
Women (15–49) Using Hygienic Menstrual Methods	0.0509***	-0.0268***	-0.0557***	-0.0337***	-0.0527***	-0.0557***	1.0000

Source: Authors' estimation based on NFHS-5 data.

Table A.3: Pairwise correlation matrix of clean cooking fuel

Variable	Type of Cooking Fuel	Ownership of Mobile Phone by Women	Account in Bank or Other Financial Institution	Type of Earnings from Women's Work	Women Participation in Decision Making	Ownership of House or Land	Women (15–49) Using Hygienic Menstrual Methods
Type of Cooking Fuel	1.0000						
Ownership of Mobile Phone by Women	-0.0151***	1.0000					
Account in Bank or Other Financial Institution	-0.0165***	0.2250***	1.0000				
Type of Earnings from Women's Work	-0.0103*	0.0332***	0.0285***	1.0000			
Women Participation in Decision Making	-0.0013	0.0690***	0.0438***	0.0191***	1.0000		
Ownership of House or Land	-0.0165***	0.2250***	1.0000***	0.0285***	0.0438***	1.0000	
Women (15–49) Using Hygienic Menstrual Methods	0.0394***	-0.0268***	-0.0557***	-0.0337***	-0.0527***	-0.0557***	1.0000

Source: Authors' estimation based on NFHS-5 data.